



TobiasBelch

PRODUCT ENGINEER

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about

Product Engineer with 17 years delivering web platforms in healthcare, e-health commerce, and sustainability. Applies product thinking to technical work: contributes to product strategy and feature definition alongside PMs and designers, then ships end-to-end. Proactively challenges over-engineered specs, grounds decisions in user and business context, and measures success by outcomes rather than output. Recognises that engineering choices are product decisions: every architecture, abstraction, and investment either accelerates or limits what the product can do next.

Drives AI workflow optimisation through reliable agent patterns, schema-validated LLM outputs, and structured prompt engineering: turning AI from unpredictable novelty into dependable product capability. Combines this with Design System initiatives as dual levers for product velocity: AI for generation speed, Design Systems for assembly consistency. Consistently sought out as a technical reference point across teams; owns stakeholder communication in high-visibility situations and mentors engineers to raise collective quality and delivery speed.

skills

Product Engineering

Product Thinking & Discovery
Cross-Functional Collaboration
Frontend Architecture
Stakeholder Communication

Frontend

TypeScript / JavaScript
React.js / Next.js
Redux / State Mgmt
Material UI / Tailwind / SCSS
D3 / Data Visualisation

Backend & API

Node.js
GraphQL / REST APIs
NestJS
Docker

Quality & Testing

Jest / Cypress / Playwright

AI & Emerging

LLM Integration / Prompt Engineering
AI Agent Orchestration
Skill Development

languages

German (Native) · English (CEFR C-level)

work experience

Senior Software Engineer – DocMorris

Berlin, Germany | 08/2023 – Present

Digital healthcare and online pharmacy platform in the German market.

- Led a white-label monolith-to-multi-tenant migration. Onboarded 3 brands across 11M active users with ~90% code reuse; second tenant launched in 2 months vs. an estimated 2 years, a ~92% time-to-market reduction, while improving testability and maintainability over the original monolith.
- Identified LLM unpredictability as a blocker for product creation workflows and established reliable agent patterns, piloted in-house, then handed to 3 additional teams. Achieved an estimated 30x token-cost reduction on knowledge-retrieval tasks with schema-validated outputs approaching deterministic reliability.
- Designed a cross-platform Design System (~30–40 components) to eliminate designer-engineer handover overhead and enable AI-assisted UI generation. Projected to halve design time and reduce implementation effort by an estimated 80%.
- Established a structured mentorship program for 5 of 15 engineers after recognising that career-growth velocity was a retention risk: combining 1:1s, pair programming, and weekly education sessions. One bootcamp-graduate engineer promoted to mid-level within 2 years and to senior within 4, a 2-year acceleration vs. industry average.
- Drove frontend and BFF architecture evolution to keep the platform adaptable as business priorities shifted. Trusted liaison for critical stakeholder communication with product and business teams.
- Championed test coverage and accessibility as product-quality investments: reducing regression risk and expanding addressable user base.

Head of Engineering – Made2Flow

Berlin, Germany | 12/2022 – 07/2023

Sustainability platform for fashion supply-chain decision support.

- Co-designed a next-generation domain data model with the CPO after identifying that the MVP-stage model was blocking ~30 business cases. Delivered the redesign in 6 months alongside other responsibilities, directly unblocking several client deals that were pending the transformation.
- Recognised that recurring friction between business goals and implementation stemmed from missing domain documentation and refinement processes; established both, closing the alignment gap and enabling the team to ship with fewer mid-development course corrections.
- Built and led a team of 4 engineers, hiring 2 within 2 months. Owned technical partner communication and integration planning to keep product delivery on track across external dependencies.

Technical Lead – eHealth-Tec GmbH

Berlin, Germany | 04/2020 – 11/2022

Digital health products for patients, doctors, and pharmacists.

- Replaced ad-hoc architecture decisions, where proposals were lost and arbitrary shifts caused chaos, with a structured ADR and design review process (~15–20 ADRs). Accelerated decision-making, improved codebase stability, and reduced onboarding time; process still active 5 years on.
- Architected BFF layers insulating the frontend (~11M users) from restructuring across ~20 backend services: eliminating ripple-effect changes and enabling white-label tenant differentiation.
- Built and owned the frontend of a white-label online pharmacy platform serving 2 clients in production (TypeScript, React, Next.js). Balanced client-specific requirements against platform reusability, a constant product-engineering trade-off.
- Led stakeholder communication and cross-functional alignment for a team of 4, bridging product goals with technical delivery.
- Mentored and onboarded engineers, growing team capability in TypeScript, architecture, and testing.
- Delivered a gematik-regulated browser-based e-prescription tool for physicians, navigating regulatory constraints without compromising usability or timeline.

work experience

Head of Engineering – Medtower UG

Berlin, Germany | 09/2018 – 04/2020

Software and hardware solutions for hospital operations and visibility.

- Architected a real-time distributed communication system using AMQP 1.0 / Apache Qpid where IoT hardware, servers, and web browsers operated as peers via pub/sub: eliminating point-to-point integration complexity. The protocol's finance-sector provenance ensured the uptime and reliability required in a hospital environment. This was the core infrastructure enabling all product use cases.
- Built 3D indoor multi-floor maps with live device location for hospital equipment tracking (beds, wheelchairs, devices), including indoor navigation and reporting views. Entire system operated in an offline hospital intranet (no internet access), requiring fully self-contained tile serving and real-time updates.
- Managed an external Apache NiFi consultant for client-specific ETL pipelines. Within 2 weeks of engagement, the output enabled the team to build business cases, demonstrating rapid time-to-value from the ETL infrastructure.
- Owned architecture and development of a real-time IoT platform: microservices, SPA, and live device tracking (Node.js, React, TypeScript).
- Led a team of 2 engineers; established requirements engineering, refinement, and team knowledge base.

earlier experience

Software Developer – remind me GmbH

02/2018 – 08/2018

Authored the technical concept for the white-label product transformation, adopted and shipped after tenure. Implemented feature-toggle architecture with lazy loading for flexible multi-context deployment. Mentored junior and mid-level developers in a Kanban team of 5.

Software Developer – Cedavis Technology GmbH

08/2009 – 01/2018

Evolved sole ownership of frontend architecture over 9 years: introduced Angular 1, then migrated to React and TypeScript as the stack matured. Built a custom D3 chart component library from scratch. Engineered a PDF reporting module using Java Nashorn to render React components server-side, enabling a single shared codebase for live UI and PDF output. Implemented data simulation and forecasting features for healthcare capacity planning.

certifications

iSAQB Certified Professional for Software Architecture – Foundation

05/2022

o0se

Domain Driven Design

12/2021

entwickler.de

education

Berlin University of Applied Sciences (BHT)

2007 – 2011

Bachelor of Science, Computer Science